

Road Transport Organization

(User, Officer (Clerk, Inspector, SuperIntendent))

US01: Register a New Driver

- As an RTO Officer (Clerk), I want to register a new driver with their personal and license details so that they can legally drive.

Business Rules:

- Driver must be at least 18 years old
- License number must be unique across all drivers
- License expiry must be at least 1 year from issue date
- Only active RTO officers can register drivers

US02: Register a New Vehicle

- As an RTO Officer (Clerk), I want to register a new vehicle under an existing driver.

Business Rules:

- Vehicle must be registered to an active driver.
- Registration number format must match state pattern (e.g., KA-01-AB-1234).
- Same vehicle cannot be registered more than once.
- Driver's license must be valid at time of registration.

US03: Record a Traffic Violation

- As an RTO Officer (Inspector), I want to record a traffic violation against a driver and vehicle.

Business Rules:

- Violation can only be recorded against active driver and vehicle
- Vehicle must belong to the cited driver
- Each violation type has predefined fine amounts
- Only officers with "Inspector" role can record violations

US04: Pay a Violation Fine

- As a Driver, I want to pay my violation fine online.

Business Rules:

- Fines become overdue after due date
- Overdue fines accrue 10% penalty
- Payment must be full amount (no partial payments)
- Paid fines cannot be modified

US05: Renew Road Tax

- As a Vehicle Owner, I want to renew my road tax before expiry.

Business Rules:

1. Tax can be renewed up to 1 month before expiry
2. Vehicle must be active to renew tax
3. Tax duration is exactly 1 year from payment date
4. Non-compliant vehicles cannot renew until all violations cleared

US06: View Driver Details

- As an RTO Officer (Inspector / Superintendent), I want to see a driver's profile, vehicles, and violations.

Business Rules:

1. Only authenticated RTO officers can view full driver details
2. Regular officers see basic info; inspectors see complete history
3. Mask license number digits for privacy (display only last 4)
4. Show warning if license expires within 30 days
5. Highlight unpaid violations in red

Special Cases:

1. Drivers with 3+ violations show "High Risk" flag

US07: View Vehicle Details

- As an RTO Officer (Inspector / Superintendent), I want to check a vehicle's compliance status.

Business Rules:

1. Display tax compliance status prominently
2. Show inspection due date (6 months from last inspection)
3. Commercial vehicles display additional cargo information
4. Highlight if involved in >3 violations in past year
5. Show recall notices from manufacturer if applicable

Data Rules:

1. Registration date must be <= first violation date
2. Cannot show vehicles marked as "stolen" to public users

US08: List all expired licenses

- As an RTO Inspector, I want to see drivers with expired licenses for follow-up.

Business Rules:

1. Sort by days expired (most recent first)
2. Exclude licenses already in renewal process
3. Show count of associated vehicles for each driver
4. Highlight licenses expired >90 days (legal action required)

Action Rules:

1. "Send Notice" button disabled for licenses expired >1 year
2. Bulk notice sending limited to 50 at a time.

US09: List all vehicles with Unpaid tax

- As an RTO Inspector, I want to identify non-compliant vehicles.

Business Rules:

1. Group by months overdue (1-3, 3-6, 6+)
2. Show accumulated late fees
3. Disable "Send Alert" for vehicles already impounded
4. Highlight vehicles with both unpaid tax and violations
5. Show owner contact information for officers ranked Inspector+

Calculation Rules:

1. Late fee = (months overdue) * 5% of original tax
2. Maximum late fee capped at 25% of original tax

US10: Dashboard for RTO Officers

- As an RTO Officer, I want a quick overview of pending tasks.

Business Rules:

1. Show different metrics based on officer role
 - Clerks: Pending registrations
 - Inspectors: Compliance rates
 - Supervisors: Revenue metrics

US11: Validate driver before registration

- **Description:** Ensure a driver exists and has a valid license before assigning a vehicle.

Business Rules:

1. Driver must exist in system
2. Driver's license must be active and not expired
3. Driver cannot have more than 5 registered vehicles

US12: Automatically mark vehicle as Non-Compliant if Tax expires

- **Description:** If road tax is not renewed, mark the vehicle as non-compliant.

Business Rules:

1. Daily check for expired taxes
2. Non-compliant vehicles cannot be driven
3. 15-day grace period before marking non-compliant
4. Notify owner when marked non-compliant.

US13: Apply late penalty for unpaid-bills.

- **Description:** If a violation fine is unpaid after the due date, add a 10% penalty.

Business Rules:

1. Penalty applies at 12:01 AM after due date
2. Maximum 3 penalties applied (30% total)

3. After 90 days, case escalates to court
4. Penalties are irreversible.

US14: Restrict Violation entry for invalid license.

- **Description:** Do not allow violations if the driver's license is expired.

Business Rules:

1. Violations cannot be added to expired licenses
2. Suspended licenses cannot receive new violations

US15: Generate monthly tax remainders

- **Description:** Send reminders to vehicle owners 15 days before tax expiry.

Business Rules:

1. First reminder at 30 days before expiry
2. Second reminder at 15 days
3. Final notice at 5 days
4. No reminders for non-compliant vehicles

US16: Validate vehicle ownership before violation entry

- **Description:** Ensure the vehicle is registered under the driver before recording a violation.

Business Rules:

1. Vehicle must be currently registered to cited driver
2. *Reject if ownership changed <24 hours before violation*

US17: Automatically deactivate expired licenses

- **Description:** If a driver's license expires, mark them as inactive.

Business Rules:

1. 15-day grace period before deactivation
2. Maintain "previously active" status for 1 year
3. Prevent new violations but allow access to history

US18: Storage Backup

- **Description:** Expired or Deactivated licenses must be archived in new database table.

Business Rules:

1. *Maintain the archive database for security audits.*

US19: Restrict Officer access based on role

- **Description:** Only "Inspector" role officers can approve license renewals.

Business Rules:

1. *Three-tier role hierarchy:*
Clerk: Data entry only
Inspector: Approvals up to ₹10,000 fines
Superintendent: Unlimited authority

US20: Generate monthly violation reports

- **Description:** Export a monthly report of all unpaid violations.

Database Schema

Drivers
Driver_id (PK)
Name
DOB (Date of Birth)

License_number (Unique)
License_issue_date
License_expiry_date
Created_on (timestamp)
Created_by
modified_on (timestamp)
modified_by
Is_active (Boolean)

Violations
Violation_id (PK)
Driver_id (FK – Drivers)
Vehicle_id(FK – Vehicles)
Violation_type
Fine_amount
Due_date
Created_on (timestamp)
Created_by
Paid_on (timestamp)
Is_paid (Boolean)

RTO_Officers
Officer_id (PK)
Name
Email
password
Badge_number (unique)
Joining_date
Role (Clerk / Supervisor / Inspector)

Vehicles
Vehicle_id (PK)
Registration_number (Unique)
Model
Manufacturer
Registration_date
Owner_id (FK – Drivers)
Created_on (timestamp)
Created_by
modified_on (timestamp)
modified_by
is_active (Boolean)

RoadTax
Tax_id (PK)
Vehicle_id (FK – Vehicles)
Amount
Issue_date
Expiry_date
Created_on (timestamp)
Created_by
modified_on (timestamp)
modified_by
is_paid (Boolean)